

Section 4: Ecology and the environment

- a) The organism in the environment
- b) Feeding relationships
- c) Cycles within ecosystems
- d) Human influences on the environment

- a) The organism in the environment

Students will be assessed on their ability to:

- 4.1 understand the terms: population, community, habitat and ecosystem
- 4.2 recall the use of quadrats to estimate the population size of an organism in two different areas
- 4.3 describe the use of quadrats as a technique for sampling the distribution of organisms in their habitats.

- b) Feeding relationships

Students will be assessed on their ability to:

- 4.4 recall the names given to different trophic levels to include producers, primary, secondary and tertiary consumers and decomposers
- 4.5 understand the concepts of food chains, food webs, pyramids of number, pyramids of biomass and pyramids of energy transfer
- 4.6 understand the transfer of substances and of energy along a food chain
- 4.7 explain why only about 10% of energy is transferred from one trophic level to the next.

- c) Cycles within ecosystems

Students will be assessed on their ability to:

- 4.8 describe the stages in the water cycle, including evaporation, transpiration, condensation and precipitation**
- 4.9 describe the stages in the carbon cycle, including respiration, photosynthesis, decomposition and combustion
- 4.10 describe the stages in the nitrogen cycle, including the roles of nitrogen fixing bacteria, decomposers, nitrifying bacteria and denitrifying bacteria (specific names of bacteria are not required).**

d) Human influences on the environment

Students will be assessed on their ability to:

- 4.11 understand the biological consequences of pollution of air by sulphur dioxide and by carbon monoxide
- 4.12 recall that water vapour, carbon dioxide, nitrous oxide, methane and CFCs are greenhouse gases
- 4.13 understand how human activities contribute to greenhouse gases
- 4.14 understand how an increase in greenhouse gases results in an enhanced greenhouse effect and that this may lead to global warming and its consequences
- 4.15 understand the biological consequences of pollution of water by sewage including increases in the number of microorganisms causing depletion of oxygen**
- 4.16 understand that eutrophication can result from leached minerals from fertiliser
- 4.17 understand the effects of deforestation, including leaching, soil erosion, disturbance of the water cycle and of the balance in atmospheric oxygen and carbon dioxide.